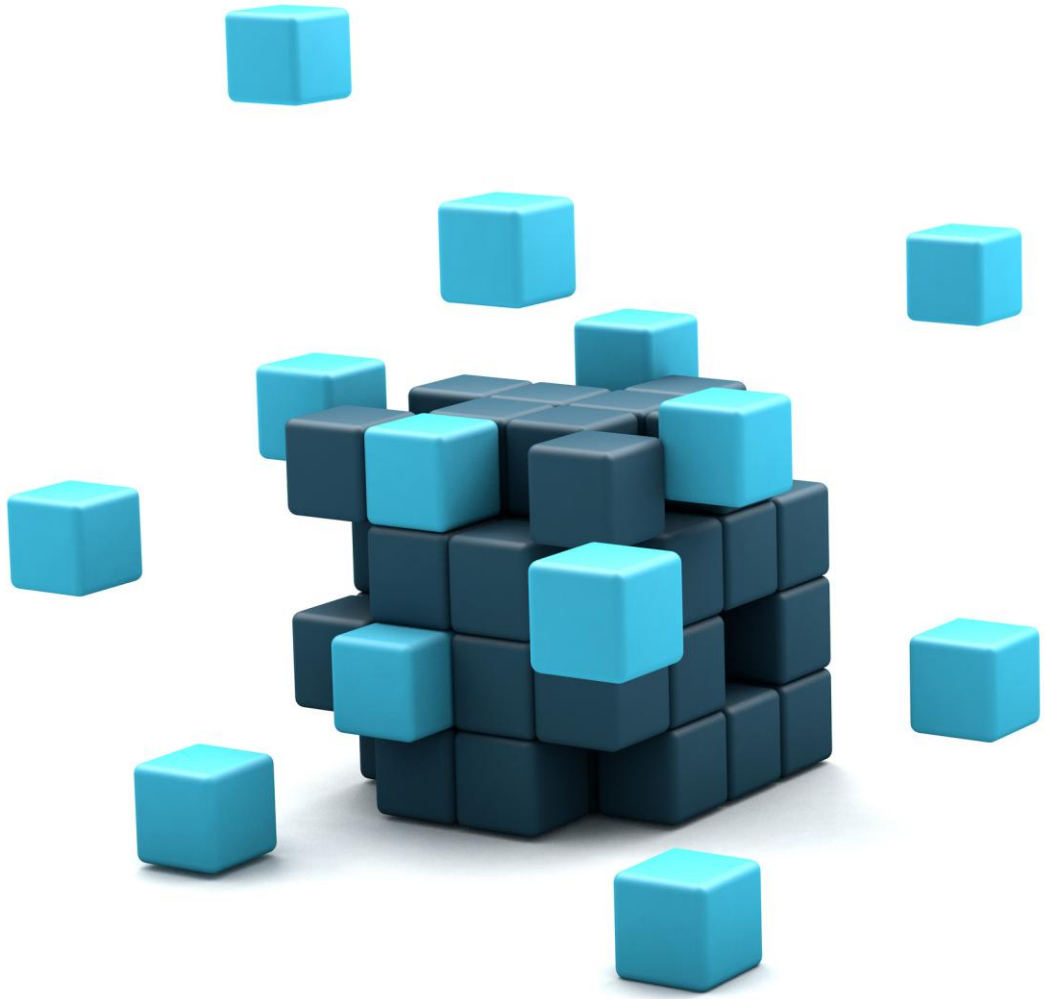
BLOCKCHAIN

EMERGING TECHNOLOGY COLLABORATION PROJECT



BLOCKCHAIN

- Fall 2024 for Professor **Carla Speights**
- By Team #3 – ITE152 Students
- Krithik Miriyala
- Nathaniel Price
- Cecilia Smith

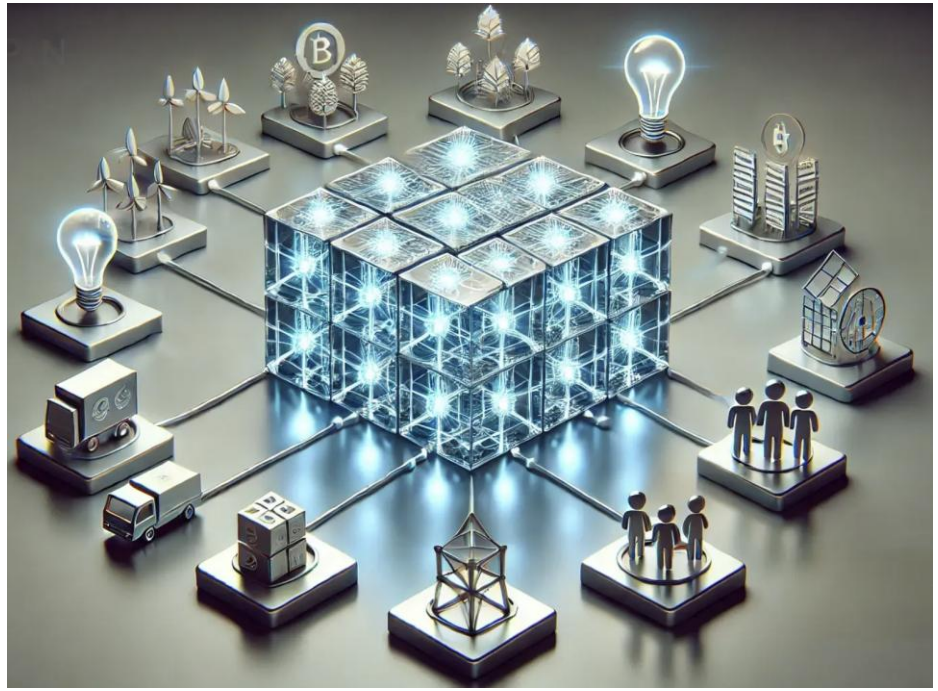
BLOCKCHAIN- AGENDA

Overview of Blockchain - Krithik Miriyala

Components of a Block in Blockchain – Nathaniel Price

Applications of Blockchain Across Industries – Cecilia Smith

BLOCKCHAIN OVERVIEW



- A distributed database or ledger that is shared over a computer network is called a blockchain. Although they have applications outside of cryptocurrencies, they are most well-known for playing a vital part in cryptocurrency systems by upholding a safe and decentralized record of transactions. Any industry can use blockchain technology to make data immutable, or unchangeable.

BLOCKCHAIN OVERVIEW 2

- You might be familiar with spreadsheets or databases. A blockchain is somewhat similar because it is a database where information is entered and stored. The key difference between a traditional database or spreadsheet and a blockchain is how the data is structured and accessed.
- The only place where trust is required is when a user or program submits data since a block cannot be altered. This reduces the need for reliable outside parties, such auditors or other people, who can make mistakes and incur additional expenses.
- Since the introduction of Bitcoin in 2009, the number of blockchain applications has increased due to the creation of smart contracts, decentralized finance (DeFi) applications, non-fungible tokens (NFTs), and other cryptocurrencies.

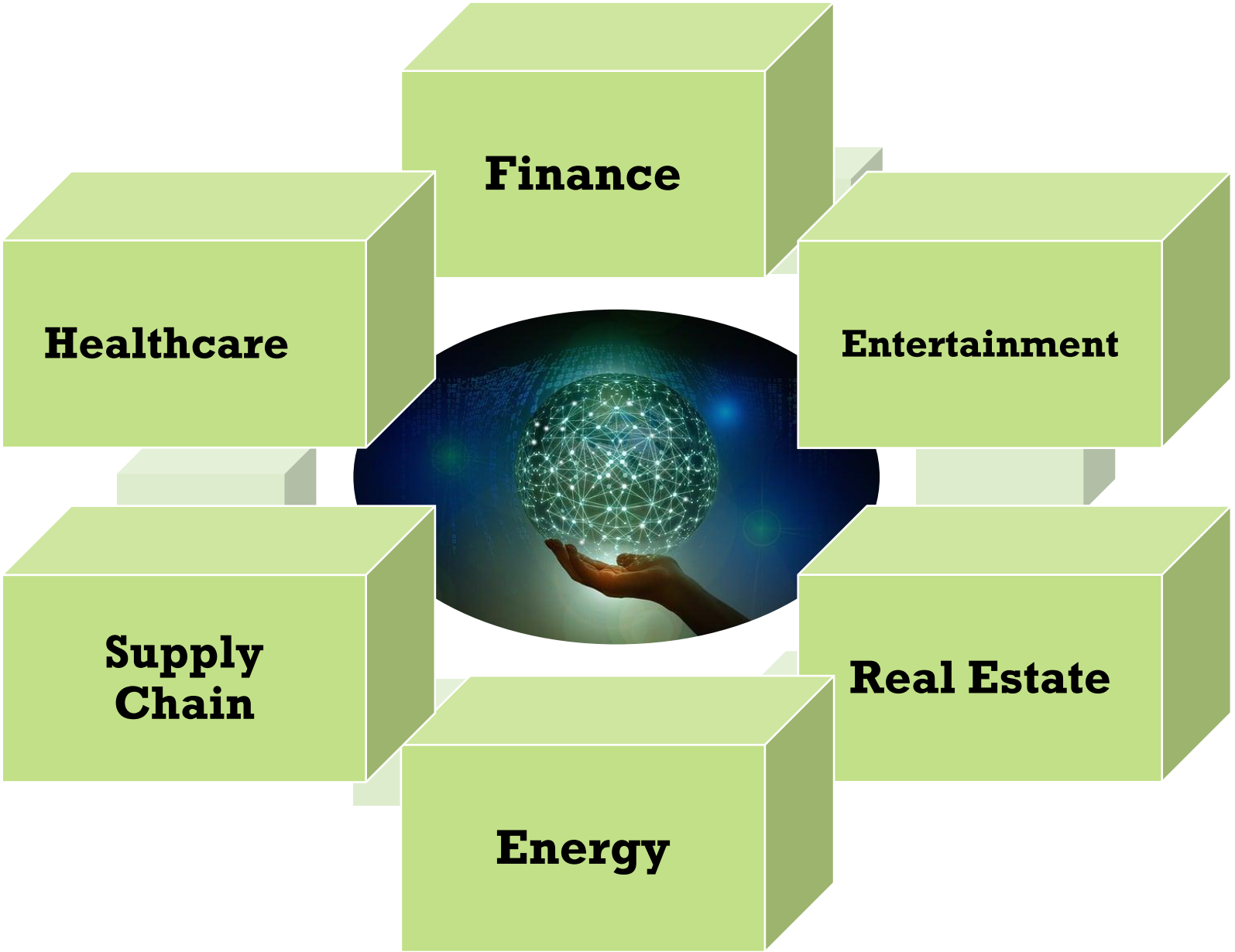
Parts of a Blockchain 1

- Nodes: Parts that participate in the blockchain network
- Ledger: Records and saves transactions in a secured manner
- Transactions: The act of data transfer between nodes and recorded by ledgers
- Consensus Mechanisms: System that keeps the blockchain working properly

PARTS OF A BLOCKCHAIN 2

- Cryptography: Secures transactions, protecting and securing data.
- Smart Contract: Self-executing contracts that run on the blockchain
- Tokens: Digital objects that represent parts of a blockchain

BLOCKCHAIN APPLICATIONS ACROSS INDUSTRIES



LESSONS LEARNED

- **Krithik Miriyala:** This group project concept was really fascinating and thrilling to work on. I learned what a blockchain is and, more importantly, how to partner with my amazing team to complete this project. I had a great time during the entire process.
- **Nathaniel Price:** Minecraft Redstone is like a blockchain, you have your signals (Levers, pressure plates, buttons), nodes (Repeaters), and your transactions (Redstone dust).

LESSONS LEARNED

- **Cecilia Smith:** Even though it was a challenge to find time to meet, especially with everyone's busy schedules, it was really rewarding to work through those communication challenges and finish the project like putting together a puzzle. Thanks to my teammates and the class – I learned a lot!

STUDENT CONTRIBUTION SUMMARY

Krithik Miriyala: Slides 4,5

Nathaniel Price: Slides 6,7

Cecilia Smith: Slides 10,11

CITATIONS

Works Cited

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THANK YOU